

LOWER MUSTANG, GANDAKI PROVINCE, NEPAL | ~2,650 m

MARPHA (MARFA)

Widely Known as Apples · Thakali heartland village · ~1,500 residents, ~414–500 households ·



GEOGRAPHY & LANDSCAPE

- Elevation: ~2,650 m a.s.l.
- Situated in the Kali Gandaki River valley, Lower Mustang, Gandaki Province
- Tucked behind a mountain ridge on the western bank of the Kali Gandaki — this shelters it from the fierce afternoon valley winds that affect open sections of the gorge
- Located roughly 7 km south of Jomsom (Mustang district headquarters) and ~30 km south of Kagbeni
- Part of the Panchgaon (5 villages) sub-region of Lower Mustang; distinct from the Baragaon (12 villages) to the north
- The Kali Gandaki Gorge here is often cited as one of the world's deepest gorges, flanked by Dhaulagiri (8,167 m) and Annapurna I (8,091 m)
- Stone-paved alleyways, flat-roofed whitewashed houses, wood stacked on rooftops — uniform Thakali vernacular architecture
- An efficient underground drainage system runs under the main lanes — unusual for a village at this altitude
- Tibetan refugee camp on the opposite (eastern) bank of the Kali Gandaki; Chhairo Gompa located nearby

CLIMATE & WEATHER

- Semi-arid, rain-shadow climate; significantly drier than southern Nepal
- Strong afternoon winds up the Kali Gandaki corridor are a near-daily feature — driven by pressure differences between the Tibetan plateau and lower valley
- Short growing season; cold winters with snowfall; high diurnal temperature variation
- Snowfall declining markedly over the past two decades — critical for apple orchards which need a cold winter dormancy period
- Rising temperatures are making Lower Mustang increasingly unsuitable for apples while simultaneously opening new apple-growing areas in Upper Mustang (as far north as Lo Manthang, 3,840 m)
- Flash flooding events in what was historically an arid rain-shadow valley — increasing reports of flash floods in recent years as monsoon patterns shift
- Untimely rainfall and cloudbursts outside the monsoon season recorded — the June 2015 event in Marpha occurred three weeks before the monsoon was due

BIODIVERSITY

Fauna

- Mule rearing historically central to Thakali economy (pack animals for salt trade routes)
- Livestock: goats, sheep, horses, yaks, dzo (yak-cattle cross) in surrounding pastoral areas
- Tibetan refugee camp near village supports livestock trade and animal husbandry
- Limited wild fauna in immediate valley; higher-altitude species in surrounding hills

Flora

- Apples: the dominant crop Red Delicious and Golden Delicious varieties; orchards support major share of the village economy
- Other crops: barley, buckwheat, millet, potatoes, kidney beans, vegetables
- Apple-derived products: wine, cider, brandy (Marpha Brandy), jam, dried apple, apple pie all sold locally and exported to Kathmandu and Pokhara
- Horticulture Development Centre (est. 1966) introduced improved apple saplings and remains active
- Pine forest patches along lower Kali Gandaki sections south of Marpha

Aquatic / Water

- Kali Gandaki River is the defining feature of the valley glacier-fed, highly seasonal
- Normally a manageable river through this section; becomes a serious hazard during monsoon and glacial lake outburst events
- Groundwater and lateral streams used for orchard irrigation
- Road construction since 2017 has destabilised slopes above the river, increasing sediment loads



Stone-paved alleyways, flat-roofed whitewashed houses, wood stacked on rooftops uniform Thakali vernacular architecture



Apple Orchard at Marpha Mustang

Landslides and Slope Instability

- Pre-historic sand and boulder geology with poor vegetation cover easily mobilised by intense rain
- June 2015: pre-monsoon cloudburst sent a wall of liquefied mud down a tributary into Marpha; 25 houses damaged (Nepali Times)
- May 2015: post-Gorkha earthquake landslide dammed the Kali Gandaki at Ranche; breach released a significant downstream flood timely evacuation of Beni prevented casualties
- 2020 monsoon: cascading multi-hazard events in three Kali Gandaki tributaries recorded by satellite and seismic data highest annual rainfall at nearby stations since 1970
- Road construction (2017 onwards) along the two-lane Kali Gandaki highway has undercut slopes and reactivated deep-seated landslides
- Road construction is believed to have contributed to slope instability
- Topsoil loss from deforested hillsides increases sediment runoff into the river

Kali Gandaki corridor

- Topsoil loss from deforested hillsides increases sediment runoff into the river

Drought and Water Scarcity

- Declining snowfall means less snowmelt for dry-season irrigation
- Paradox of drought in dry season and flooding in monsoon both increasingly extreme
- Farmers report inability to irrigate orchards adequately in spring

Out-migration

- Long-term depopulation as Thakali families moved to Pokhara and Kathmandu following salt trade decline
- Continuing as younger generations leave for education and urban employment
- Reduces agricultural and cultural workforce in the village

Extreme Cold

- Hard winters limit agricultural activity and access to healthcare
- Health sub-post exists in Marpha (one of 17 in Mustang district) but services are limited

Agricultural Hazards — Apple Crop Decline

- Invasive pest species spreading with warming: red spider mite, woolly aphid, white grub
- Insufficient winter snowfall leaves orchards without cold dormancy — reducing fruit set
- 2023: apple production in parts of Mustang dropped ~50% due to drought and locust attacks
- Apple growing zone is effectively migrating uphill — Marpha's lower altitude becoming too warm

LIVELIHOODS & AGRICULTURE

- Historically: trans-Himalayan salt trade was the economic foundation Thakali traders acted as intermediaries between Tibet and southern Nepal
- After salt trade collapse (post-1959): shift to apple cultivation, mule rearing, and tourism hospitality
- Apple orchards: primary income for most families; products (brandy, wine, jam, dried fruit) sold across Nepal
- Tourism: Marpha is a standard Annapurna Circuit overnight halt; guesthouses and teahouses run largely by Thakali families
- Remittances and diaspora income increasingly important as local agriculture comes under climate stress
- Some families can no longer cover orchard rent from apple sales alone income squeeze documented by Kathmandu Post (2024)

CULTURE, RELIGION & INDIGENOUS KNOWLEDGE

- Nyingma Tibetan Buddhism is the primary religious tradition; Samten Choling Monastery (on cliff above village) is the main site — evening puja open to visitors
- Chhairo Gompa across the Kali Gandaki: a 16th-century Nyingma monastery, said to be the first Nyingma monastery established in Mustang
- Mamti cave with a bronze statue of Padmasambhava; stone-carved Tibetan texts on surrounding slates
- Rhisum Gompa: a triangular white chorten on the hillside north of the village
- Chorten-building tradition: villagers believe constructing chortens and holding rituals at them averts illness and disaster
- Traditional firewood stacking on rooftops: a practical necessity and a marker of household wealth
- Festivals: Losar (Tibetan New Year, Feb–Mar) and Mani Rimdu (autumn) are main communal celebrations
- Thakali community governance system historically notable for its flexibility and adaptive management of common resources

KEY HISTORICAL EVENTS

Period	Event	Theme
Pre-19th c.	Active trans-Himalayan salt trade; Thakali traded between Tibet and Nepal	
Mid-19th c.	Thakali leaders collaborate with Rana regime; granted control of rock-salt import from Tibet	
Pre-1960s	Peak of Tibetan salt trade along Kali Gandaki; Marpha a key waypoint	Economy
1959–1960s	Chinese annexation of Tibet and stricter border controls disrupt trade; Thakali out-migration	
1966	Horticulture Development Centre established; apple cultivation introduced and expanded	Agriculture
1990s–2000s	Annapurna Circuit opens to mass trekking; Marpha becomes a regular overnight halt	Tourism
2015 (May)	Post-Gorkha earthquake landslide dams Kali Gandaki; dam breach flood affects valley	
2017–present	Two-lane road construction along Kali Gandaki valley increases slope instability and landslides	
2020-2025	Snowless winters and rising temperatures reduce apple yields; pests (red spider mite, woolly	

SOURCES

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